WORK IN PROGRESS

AI-Assisted Cryptographic Control Plane

Project Evidence Pack

Product requirements, architecture, detailed design, assurance, roadmap and reproducible evidence

Version 0.4.0

Milestones 1-4 complete

11 October 2026

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This combined pack records the current project position. Implemented, partial, planned and out-of-scope capabilities are deliberately separated.

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AI-Assisted Cryptographic Control Plane

Project progress summary

Project status: Work in progress **Current version:** 0.4.0 **Current milestone:** Milestone 4 completed **Last updated:** 11 October 2026

Project overview

This project is developing an **evidence-backed cryptographic discovery and post-quantum cryptography (PQC) migration control plane**.

Its purpose is to help organisations discover where cryptography is being used, record the supporting evidence, identify cryptographic components that may be affected by quantum computing, and manage future migration work in a controlled and auditable way.

Certificate monitoring is the first working discovery capability. The intended product is broader and will eventually include cryptographic inventory, dependencies, ownership, migration planning, approvals, exceptions, and verification.

Work completed

Milestone 1 - Certificate monitoring foundation

- Created a versioned certificate-event format.
- Added strict validation and DNS-name normalisation.
- Added deterministic detection of certificates containing unapproved DNS namespaces.
- Added risk scoring, evidence-based explanations, alert output, and a SOC playbook.

Milestone 2 - Evidence intake and replay protection

- Added a controlled Certificate Transparency fixture collector.
- Added evidence provenance and SHA-256 content digests.
- Added SQLite storage for observations, outcomes, and checkpoints.
- Added duplicate detection, restart support, and fail-closed provenance conflict handling.

Milestone 3 - Cryptographic inventory and PQC assessment

- Added cryptographic inventory records linked to source evidence.
- Added deterministic identification of quantum-vulnerable RSA and elliptic-curve certificate evidence.
- Added support for representing ML-KEM, ML-DSA, SLH-DSA, hybrid, and unknown algorithm evidence.
- Added explicit handling of missing ownership, business criticality, data classification, and protection-lifetime information.

- Prevented PQC algorithm names from being treated as proof that a deployment is

quantum-safe.

Milestone 4 - Controlled discovery laboratory

- Added generation of real RSA and elliptic-curve X.509 test certificates.
- Added PEM and DER certificate parsing.
- Added discovery of certificate fingerprints, identities, validity periods,

public-key algorithms, key sizes, curves, signatures, key usages, and extended key usages.

- Added a bounded loopback TLS test that records the TLS version and cipher

suite without scanning external systems.

- Added evidence-maturity levels that distinguish simulated, generated,

sanitised, public-observation, authenticated-production, and independently verified evidence.

- Added controls preventing laboratory evidence from being presented as

authenticated production evidence.

Current evidence

- Four incremental implementation milestones are recorded in Git history.
- The repository contains versioned JSON schemas, architecture decisions, a

threat model, examples, command-line demonstrations, and automated tests.

- The current automated suite contains **46 tests**.
- The suite has passed on Python 3.12 and Python 3.13, including the real local

TLS handshake test when run outside the restricted test sandbox.

- The current working implementation is recorded in commit `be71182`.
- The project documentation pack is recorded in commit `fd2d7ea`.

Current limitations

The project is a local reference implementation and remains under development. It does not currently:

- connect to customer or production data sources;
- claim complete discovery of an organisation's cryptographic estate;
- provide authenticated production collectors;
- discover every form of cryptography used in applications, code, KMS, HSM,

private PKI, cloud services, or supplier products;

- operate as a multi-user or multi-tenant production service;
- execute certificate, key, trust-store, or production migration changes;
- prove that an organisation is quantum-safe;
- carry independent certification, audit, or penetration-test assurance.

These limitations are intentional at the current stage. The project is proving the data contracts, evidence boundaries, deterministic assessment rules, safety controls, and migration-management design before production integrations or

change authority are introduced.

Next planned milestone

Milestone 5 - PQC migration management will add a fictional enterprise catalogue so that the migration workflow can be developed without customer data. Planned capabilities include:

- assets, services, owners, suppliers, and business context;
- cryptographic dependency relationships;
- current and target cryptographic states;
- migration priority and waves;
- blockers and prerequisites;
- exceptions and approval records;
- verification criteria and post-change evidence;
- separate reporting of discovery coverage and migration completion.

Overall position

The project has progressed beyond a concept or presentation. It has executable code, real cryptographic test artifacts, machine-readable contracts, persistent evidence handling, documented security boundaries, and repeatable automated tests.

It should presently be described as a **working reference implementation under active development**, not as a finished production product. The completed work provides the technical foundation for building and demonstrating PQC migration management in the next milestone.

Product Requirements Document

1. Document control

Field	Value
Product	AI-assisted Cryptographic Control Plane
Product version	0.4.0 reference implementation
Document status	Baseline for review
Date	2026-10-11
Product owner	To be assigned
Technical owner	Repository owner
Implementation status	Milestones 1-4 implemented locally

2. Product statement

The product is an evidence-backed cryptographic discovery and PQC migration control plane. It collects bounded observations, normalises them into portable contracts, creates a cryptographic inventory, classifies exposure using deterministic policy, and will govern migration work through accountable human decisions and verifiable completion evidence.

Certificate monitoring is the first sensor and reference vertical slice. It is not the final product boundary.

3. Problem

PQC migration is not only an algorithm-replacement exercise. Organisations need to know:

- where cryptography is used;
- which function each primitive performs;
- which assets, services, owners, data, suppliers, and relying parties depend

on it;

- how reliable each observation is and what it does not prove;
- which systems require earlier action because protected data must remain

confidential for a long period;

- whether products and protocols support an approved target state;
- how changes are approved, verified, rolled back, and evidenced.

Without those controls, an organisation can produce an inventory yet still be unable to prioritise or prove a migration.

4. Standards context

The design uses the functional distinction established by the final NIST PQC standards:

- FIPS 203 specifies ML-KEM for key establishment:

<https://csrc.nist.gov/pubs/fips/203/final>

- FIPS 204 specifies ML-DSA for digital signatures:

<https://csrc.nist.gov/pubs/fips/204/final>

- FIPS 205 specifies SLH-DSA for digital signatures:

<https://csrc.nist.gov/pubs/fips/205/final>

NIST's PQC programme states that organisations should identify vulnerable algorithm use and begin migration planning: <https://csrc.nist.gov/projects/post-quantum-cryptography>

NIST IR 8547 is transition guidance and remains identified as an initial public draft at this document date. Its proposed dates must therefore be treated as planning inputs, not silently converted into universal legal deadlines: <https://csrc.nist.gov/pubs/ir/8547/ipd>

5. Users and stakeholders

Persona	Need
Cryptography or PKI lead	Evidence of algorithms, certificates, protocols, trust relationships, and migration state
Security architect	Dependency and target-state views with explicit assumptions
Asset or service owner	Assigned, understandable migration work and acceptance criteria
Risk and compliance reviewer	Traceable decisions, exceptions, approvals, evidence, and coverage gaps
SOC analyst	Actionable certificate anomalies with source evidence and playbooks
Platform or application engineer	Implementation-specific compatibility requirements and test results
Executive sponsor	Exposure, coverage, blockers, forecast, and residual-risk reporting
Auditor or independent reviewer	Immutable provenance, change history, control evidence, and reproducibility

6. Goals

1 Build an evidence-backed, continuously reconcilable cryptographic inventory.

2 Preserve the distinction between observed fact, inference, recommendation, approval, execution, and verified outcome.

1 Prioritise PQC work using cryptographic exposure, data lifetime, business criticality, dependency, supplier, and readiness evidence.

1 Manage migration waves, blockers, exceptions, approvals, rollback criteria, and verification evidence.

1 Make coverage and unknowns visible rather than treating missing evidence as safety.

1 Keep AI advisory and bounded; deterministic policy and accountable humans retain authority for material decisions.

7. Non-goals

The current product is not:

- a certificate authority, KMS, HSM, secret manager, or trust store;
- an autonomous certificate or key lifecycle executor;
- a vulnerability scanner authorised to probe arbitrary networks;
- proof that any real organisation is quantum-safe;
- a replacement for protocol-specific interoperability testing;
- a compliance certification engine;
- a guarantee that every cryptographic use can be discovered automatically.

8. Product scope by capability

Capability	Current state	Target state
Canonical certificate observations	Implemented	Multiple authenticated collectors
Schema and domain validation	Implemented	Contract compatibility and producer conformance programme
Certificate anomaly detection	Implemented	Governed rule catalogue and production SIEM delivery
Replay-safe fixture intake	Implemented	Distributed, authenticated intake with transactional delivery
Cryptographic inventory	Partially implemented	Multi-source CBOM with reconciliation and lineage
PQC exposure assessment	Partially implemented	Policy-versioned portfolio prioritisation
Real X.509 parsing	Implemented for controlled files	Sandboxed production parsing and chain verification
TLS evidence	Implemented for loopback lab	Authorised scoped collectors with protocol evidence
Dependency graph	Planned	Versioned service, protocol, supplier, and relying-party graph
Migration workflow	Planned	Waves, tasks, approvals, exceptions, rollback, verification
AI assistance	Deterministic explanation only	Evidence-grounded recommendation with evaluation and guardrails
Execution	Out of scope now	Optional isolated executor after governance gates

9. Functional requirements

Discovery and evidence

- **FR-001:** Accept only versioned, validated observation contracts.
- **FR-002:** Preserve source identity, collection time, payload digest, verification state, evidence level, and cursor where applicable.
- **FR-003:** Reject replay conflicts where a source identity is reused with different content.
- **FR-004:** Parse X.509 evidence to derive, not merely accept, algorithm and certificate properties.
- **FR-005:** Keep certificate, transport, code, key-store, and configuration observations distinct until correlation is supported by evidence.
- **FR-006:** Attach explicit scope and coverage limitations to inventory data.
- **FR-007:** Support future collectors for CT, TLS, cloud certificate managers, private PKI, KMS/HSM, code/configuration, and supplier evidence.

Inventory and assessment

- **FR-010:** Create stable inventory records linked to their observations.
- **FR-011:** Represent cryptographic function separately from algorithm name.
- **FR-012:** Classify RSA, elliptic-curve, and finite-field public-key evidence as quantum-vulnerable for PQC planning.
- **FR-013:** Treat ML-KEM, ML-DSA, SLH-DSA, and hybrid labels as candidates requiring deployment and interoperability verification.
- **FR-014:** Preserve unknown values and prevent them from becoming safe defaults.
- **FR-015:** Block final migration priority when required business context is missing.
- **FR-016:** Correlate assets, services, owners, data lifetime, suppliers, dependencies, and cryptographic uses in the target product.

Migration management

- **FR-020:** Define an approved target state per cryptographic function and protocol, not one universal replacement algorithm.
- **FR-021:** Group work into migration waves based on exposure, urgency, dependency order, readiness, and change risk.
- **FR-022:** Record owner, accountable approver, planned dates, blockers, dependencies, rollback criteria, and verification criteria.
- **FR-023:** Manage time-bounded exceptions with rationale, compensating controls, expiry, and named approval.
- **FR-024:** Require pre-change approval and post-change verification evidence for material migrations.
- **FR-025:** Reconcile post-change discovery against the approved target and detect regression.
- **FR-026:** Report coverage separately from migration completion.

Detection and assistance

- **FR-030:** Evaluate security-critical detections and priority gates using deterministic, versioned policy.
- **FR-031:** Produce explanations linked to evidence and policy versions.
- **FR-032:** Prevent an AI component from silently changing evidence, severity, approval, or execution state.
- **FR-033:** Route alerts and workflow events to controlled downstream systems with delivery evidence in the target product.

10. Non-functional requirements

- **NFR-001 Security:** Fail closed for invalid contracts and unsupported evidence claims.
- **NFR-002 Auditability:** Every material conclusion must identify its input evidence, rule or model version, and decision actor.
- **NFR-003 Privacy:** Minimise, classify, redact, encrypt, and retain sensitive infrastructure metadata according to policy.
- **NFR-004 Availability:** The future service must define recovery objectives, backpressure, retry, and degradation modes before production use.
- **NFR-005 Isolation:** Collection, analysis, approval, and execution must have separate trust and privilege boundaries.
- **NFR-006 Portability:** External contracts use versioned JSON Schema; core decisions must not depend on a single vendor's data model.
- **NFR-007 Explainability:** Assessment outcomes must be reproducible without access to an LLM.
- **NFR-008 Testability:** Each mandatory requirement needs an objective acceptance test or an explicitly recorded evidence gap.
- **NFR-009 Scalability:** The target architecture must partition by tenant and source and support asynchronous processing; SQLite is not the scale design.
- **NFR-010 Accessibility:** Human workflows and reports must be usable without requiring cryptography-specialist knowledge for every field.

11. Success measures

The measures must never combine coverage and migration into one misleading percentage.

Measure	Definition
Discovery coverage	In-scope assets with at least one current observation divided by known in-scope assets
Context completeness	Inventory records with required owner, service, criticality, classification, and lifetime fields
Evidence maturity	Distribution of records by evidence level and verification state
Exposure backlog	In-scope quantum-vulnerable uses not yet assigned to an approved wave
Dependency readiness	Migration items whose blocking dependencies have an accepted target and tested path
Migration verified	Items with approved change, required tests, post-change evidence, and no unresolved regression
Exception health	Active, expired, and overdue exceptions by owner and risk
Detection quality	Validated true/false positive rates by rule and evidence source

No production target is set until an authorised pilot establishes the asset denominator, data quality, operating model, and acceptable error rates.

12. Acceptance criteria for the current reference release

1. All published example payloads conform to their schemas.
2. Invalid or unexpected event fields fail closed.
3. DNS matching is normalised and restricted to label boundaries.
4. Alert identifiers and PQC assessments are deterministic.
5. Fixture replay is idempotent and changed-content conflicts fail closed.
6. Source evidence, provenance, inventory, and assessment are linked.
7. Unknown algorithm evidence is never labelled quantum-resistant.
8. Generated X.509 certificates are parsed for RSA/ECDSA properties.
9. The lab TLS probe refuses non-loopback targets.
10. Transport evidence does not invent negotiated key establishment.
11. Fixtures and generated evidence cannot claim production assurance.
12. The complete automated test suite passes on the supported Python versions.

13. Dependencies and assumptions

- Python 3.11 or later.
- Pinned `jsonschema[format-nongpl]` and `cryptography` dependencies.
- The current repository operates locally and has no production credentials.
- Future production use assumes legal authorisation, collector identity, authoritative scope, secure hosting, operational ownership, and incident response.

14. Open product decisions

1. First authorised pilot source and accountable data owner.
2. Tenant and identity model.
3. Authoritative asset and ownership sources.
4. Workflow system of record and separation-of-duties model.
5. Policy ownership and exception authority.
6. Evidence retention, residency, and redaction rules.
7. Deployment model and service-level objectives.
8. Whether execution will remain external or use an isolated adapter.

High-Level Design

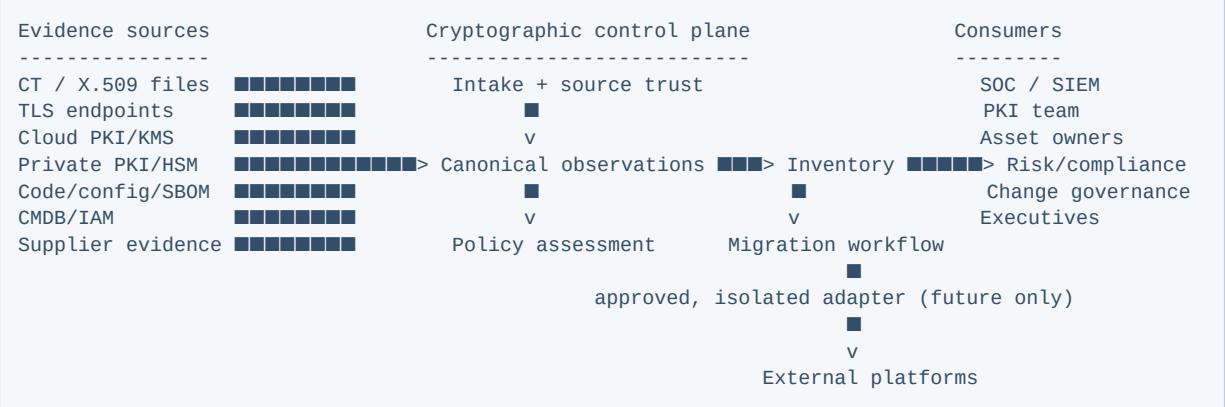
1. Architectural intent

The system is a control plane for evidence, decisions, and migration governance. It is not itself a cryptographic data plane. Applications, PKI platforms, TLS stacks, KMS/HSM products, and cloud services continue to perform cryptographic operations.

The architecture follows five rules:

1. Evidence precedes conclusions.
2. Missing evidence remains visible.
3. Deterministic policy owns security-critical decisions.
4. Approval and execution are separate trust boundaries.
5. Migration completion requires post-change verification.

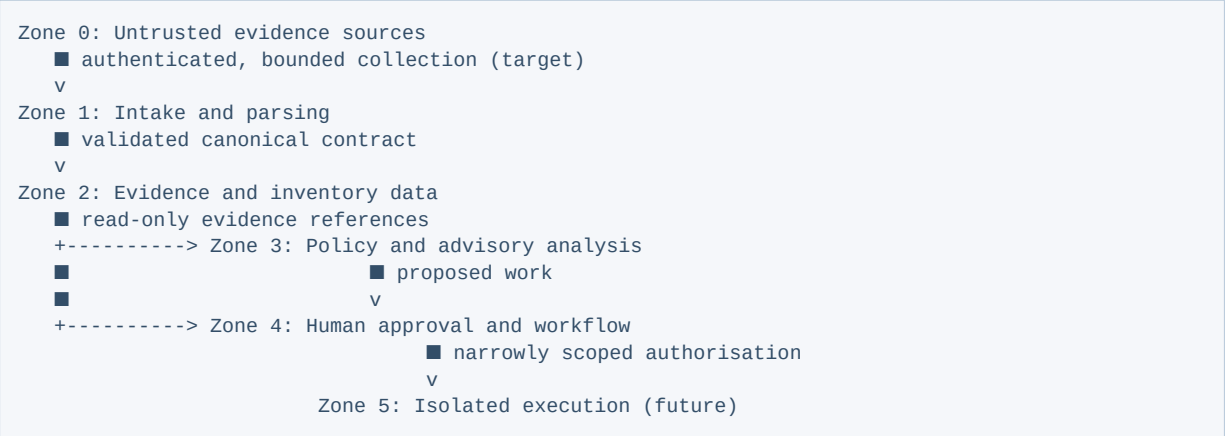
2. Context



3. Logical components

Component	Responsibility	Current status
Collector adapters	Acquire bounded evidence from an authorised source	Fixture, file, and loopback lab implemented
Intake gateway	Authenticate source, enforce limits, validate contract, attach receipt metadata	Contract validation implemented; authentication planned
Evidence ledger	Preserve raw observation, digest, provenance, outcome, and lineage	SQLite reference implementation
Normalisation service	Convert source-specific data into canonical observations	Certificate path implemented
Inventory service	Produce cryptographic-use records with coverage limits	Certificate slice implemented
Correlation and graph service	Link assets, services, identities, dependencies, suppliers, and relying parties	Planned
Policy and assessment service	Classify exposure, missing context, priority gates, and policy findings	Deterministic certificate/PQC slice implemented
Detection service	Produce operational alerts and playbook links	CERT-001 implemented
AI advisory service	Summarise evidence, suggest questions or plans, never silently decide	Deterministic fallback only
Migration workflow	Manage target state, waves, owners, approvals, exceptions, and verification	Planned
Reporting and API	Expose coverage, evidence maturity, exposure, work, and audit history	Planned
Execution adapter	Apply narrowly approved changes with rollback and evidence	Out of scope until governance gates pass

4. Trust zones



The current implementation occupies local equivalents of Zones 1-3 and has no production authority.

5. Principal data flows

5.1 Observation to inventory

1. Collector reads an authorised, bounded source.
2. Intake validates source envelope and payload schema.
3. Evidence identity and canonical digest are calculated.
4. Replay is classified as new, duplicate, or conflict.
5. Source-specific evidence is normalised without erasing provenance.
6. Inventory record and assessment are created.
7. Observation, outcome, inventory, and assessment commit atomically.

5.2 Observation to alert

1. Canonical certificate event is validated.
2. DNS identities are normalised.
3. Versioned policy evaluates the observation.
4. Deterministic severity, score, and alert identifier are created.
5. Advisory explanation is attached without decision authority.
6. Alert is validated and delivered to a sink.

5.3 Inventory to migration work (target)

1. Evidence is reconciled with authoritative asset and ownership sources.
2. Dependency paths and cryptographic functions are resolved.
3. Policy proposes priority and identifies missing evidence.
4. Owner selects a supported target state and verification plan.
5. Approvers accept the work, risk, exception, or deferral.
6. Change occurs in an external controlled delivery process.
7. Pre/post tests and new discovery evidence verify the outcome.
8. Coverage, residual risk, and regression status are recalculated.

6. Data domains

The target model has separately governed domains:

- **Observation:** what a source reported or a parser derived.
- **Evidence:** provenance, maturity, verification, integrity, scope, and time.
- **Inventory:** a cryptographic use linked to an asset or service.
- **Dependency:** a directional relationship that can block or constrain change.
- **Policy finding:** deterministic evaluation against a versioned policy.
- **Recommendation:** advisory output that requires review.
- **Migration item:** governed work with target, owner, dates, dependencies, and verification criteria.
- **Decision:** approval, rejection, exception, or risk acceptance by an actor.
- **Verification:** evidence that an approved result was achieved.

These domains must not be collapsed into a single mutable status record.

7. Deployment view

Current reference deployment

- Local Python CLI processes.
- Local JSON fixtures and generated certificates.
- SQLite evidence state.
- JSONL/file output.
- Loopback-only TLS connection.
- No credentials, tenant boundary, public API, or production service.

Target production deployment

- Collectors isolated by source and network scope.
- Authenticated ingestion endpoint with tenant binding and quotas.
- Sandboxed parsers for hostile artifacts.
- Durable event stream and immutable evidence/object storage.
- Transactional inventory and workflow stores.
- Separate policy, advisory, and approval services.
- Identity-aware API and user interface.
- Tamper-evident audit export and monitored SIEM integration.
- Optional execution adapters isolated by environment and privilege.

No cloud or platform choice is fixed by this HLD; selection requires scale, residency, integration, cost, and operating-model requirements.

8. Availability and recovery concept

Production readiness requires:

- per-source checkpoints and replay-safe ingestion;
- retry with bounded backoff and dead-letter handling;
- transactional outbox for downstream delivery;
- backup and point-in-time recovery for policy, inventory, workflow, and audit;
- evidence-object integrity checks;
- documented recovery time and recovery point objectives;
- degraded operation that stops execution before it loses auditability.

The local SQLite store demonstrates semantics, not these service levels.

9. Security architecture

- Source authentication and tenant binding occur before high-assurance evidence

labels are accepted.

- Parser processes receive no cloud or cryptographic-administration credentials.
- Evidence is append-oriented; corrections are new linked assertions.
- Policy changes use review, versioning, effective dates, and rollback.
- AI outputs are untrusted content with structured validation and citations.
- Human approval is identity-bound and separated from request creation.
- Execution credentials are narrow, expiring, environment-scoped, and absent

from analysis services.

- Sensitive infrastructure metadata is encrypted and field-authorised.

10. Key architecture decisions

1. Contract-first deterministic monitoring slice - ADR-001.
2. Replay-safe evidence intake before live collection - ADR-002.
3. Inventory and PQC assessment remain separate artifacts - ADR-003.
4. Real controlled discovery without misrepresenting lab data - ADR-004.
5. Proposed next ADR: migration-work-item state machine and decision authority.
6. Proposed next ADR: authenticated collector identity and evidence signing.

11. Architecture limitations

The current architecture does not prove complete discovery, chain validation, runtime KEM negotiation, application compatibility, supplier readiness, scale, tenant isolation, service availability, or production security. Those are explicit future validation gates.

Low-Level Design

1. Scope

This LLD documents the implemented version 0.4.0 reference slice and marks future interfaces as proposed. It should be updated whenever a schema version, security invariant, persistence rule, or public command changes.

2. Package structure

Module	Implemented responsibility
`models.py`	Strict domain parsing for certificate events and monitoring policy; evidence-claim enforcement
`schema_validation.py`	Cached JSON Schema Draft 2020-12 validation with format checking
`validation.py`	Exact-key, string, UUID, timestamp, and list guards
`normalization.py`	DNS/IDNA normalisation and policy matching
`pipeline.py`	CERT-001 enrichment, deterministic alert decision, severity, score, and explanation
`ct_collector.py`	Bounded decoded-fixture pages, provenance, replay, checkpoints, inventory creation
`state.py`	SQLite schema and atomic outcome/inventory persistence
`inventory.py`	Inventory projection and deterministic PQC exposure assessment
`discovery_lab.py`	Synthetic X.509 generation, PEM/DER parsing, and loopback TLS observation
`sinks.py`	JSONL development alert sink
`*_cli.py`	File-oriented command entry points

3. Contract catalogue

Contract	Version	Producer/consumer
Canonical certificate event	1.3	Collectors and parsers / pipeline and inventory
CT observation page	1.1	Fixture source / CT collector
Monitoring policy	1.1	Policy owner / monitoring pipeline
Detection alert	1.2	Pipeline / JSONL or future SIEM adapter
Cryptographic inventory record	1.1	Inventory service / state and future graph
PQC assessment	1.1	Inventory service / future workflow and reporting

Contracts reject unexpected fields. A producer must make an explicit version upgrade; the implementation does not silently reinterpret older events.

4. Certificate event processing

4.1 Validation sequence

```
JSON payload
  -> full JSON Schema validation
  -> exact domain keys and enum checks
  -> timestamp and certificate validity ordering
  -> algorithm family and parameter checks
  -> context validation
  -> provenance and evidence-level policy
  -> optional transport validation
  -> immutable CertificateEvent
```

Material constraints include:

- SHA-256 fingerprints and payload digests are 64 hexadecimal characters.
- Timestamps must include timezone information.
- `not_after` must be later than `not_before`.
- X.509 subject may be empty; SAN may carry identity under RFC 5280 semantics.
- KEM families are not valid certificate-signature families.
- Transport key establishment may be `null`; no value is inferred from a

certificate or cipher-suite label.

4.2 Evidence-claim policy

Collector	Allowed evidence level
`fixture-generator`	`simulated`
`ct-decoded-fixture`	`simulated`
`x509-file-parser`	`generated`, `sanitised-capture`
`tls-loopback-lab`	`generated`

`authenticated-production` and `independently-verified` exist in portable schemas but are rejected at runtime until a trusted collector registry exists. An unknown collector cannot self-assert either level.

5. Monitoring rule design

CERT-001: unknown DNS namespace

Inputs:

- certificate subject and SAN DNS names;
- environment;
- monitoring policy exact names and allowed suffixes.

Processing:

1. Lowercase and remove a final DNS dot.
2. Convert Unicode names using IDNA.
3. Permit a wildcard only as the complete leftmost label.
4. Compare suffixes only at DNS label boundaries.
5. Emit no alert when every name is covered.
6. Otherwise assign `high` in production and `medium` elsewhere.
7. Calculate score as a deterministic base plus a bounded increment for

additional unmatched names.

1. Generate a UUIDv5 alert identifier from event, rule, policy ID, and policy

version.

The result identifies a policy mismatch, not malicious issuance. The SOC playbook requires evidence and ownership verification.

6. CT fixture collection

6.1 Bounds

The fixture adapter applies file/page and entry-size limits before downstream processing. A future live adapter must separately define authenticated transport, log trust, pagination, proof verification, timeouts, download limits, and backoff.

6.2 Replay algorithm

For each source record:

1. Canonicalise raw JSON bytes.
2. Calculate SHA-256.
3. Query `(collector_id, source_record_id)`.
4. If absent, process and store.
5. If the stored digest matches, classify as duplicate and emit no second

alert.

1. If it differs, raise a provenance conflict and stop.
2. Advance the page checkpoint only after every page entry has an outcome.

Rejected domain events are preserved as rejected evidence. Malformed pages fail before state mutation.

7. Persistence model

The SQLite reference store maintains:

- collector checkpoint by collector ID;
- observation identity and raw/canonical payload;
- payload digest and processing status;
- canonical event or validation error;
- alert, inventory record, and assessment where produced;
- processing timestamp.

Observation outcome, inventory record, and assessment are committed in one transaction. JSONL delivery is not atomic with SQLite; production requires a transactional outbox.

8. Inventory projection

For one certificate event, `PqcInventoryService` creates a stable UUIDv5 record for the X.509 public-key use. It preserves:

- observation identity and time;
- asset/service/business context without inventing missing values;
- function, protocol, implementation placeholders, and algorithm;
- certificate identity, validity, usages, and signature algorithm;
- optional transport observation;
- evidence confidence, evidence level, source evidence, and event provenance;
- a `certificate-only` scope and explicit limitations.

The current projection does not merge multiple observations into a golden asset record. That requires correlation policy and conflict lineage.

9. PQC assessment algorithm

Inputs are the public-key family, certificate-signature family, inventory context, and evidence. The algorithm is deterministic:

Condition	Exposure result
RSA, elliptic-curve, or finite-field DH is observed	`quantum-vulnerable`
Any relevant family is unknown/other and no known vulnerable family exists	`unknown`
Hybrid is observed without deployment proof	`hybrid-unverified`
Only recognised PQC families are observed	`candidate-quantum-resistant`

Known vulnerability dominates unknown evidence. Recognised PQC names do not prove chain, protocol, implementation, peer, or fallback readiness. Missing business context is returned and blocks final priority.

10. X.509 discovery laboratory

10.1 Synthetic factory

The factory can generate:

- RSA 2048, 3072, and 4096-bit certificates;
- ECDSA P-256 and P-384 certificates;
- valid, expired, and not-yet-valid periods;
- CA and end-entity profiles;
- SAN and no-SAN cases.

Keys are synthetic test material. They must never be imported into a production store.

10.2 File parser

The parser:

1. Rejects files larger than one megabyte before parsing.
2. Accepts PEM or DER.
3. Uses PyCA `cryptography` to derive certificate properties.
4. Maps public-key and signature OIDs into the canonical algorithm model.
5. Extracts key usage, EKU, SAN, validity, issuer, subject, serial, and SHA-256

fingerprint.

1. Does not place an absolute local file path into source evidence.
2. Validates the resulting canonical event before returning it.

10.3 Loopback TLS probe

The probe:

- accepts an IP literal, not a host name;
- requires the IP to be loopback;
- limits timeout to ten seconds;
- performs a local TLS client handshake;
- records TLS version and cipher suite;
- explicitly records that the peer was not identity-verified;
- leaves negotiated key establishment `null` because the selected Python API

does not expose sufficient direct evidence.

11. Command interfaces

Command	Required inputs	Output
`crypto-control-plane`	event, policy, output	Zero or one JSONL alert
`crypto-control-plane-ct-fixture`	fixture directory, policy, state, output	Collection counts, SQLite evidence, alerts
`crypto-control-plane-inventory`	event, output	Inventory and assessment JSON bundle
`crypto-control-plane-x509-lab`	certificate, evidence level, output	Parsed event, inventory, assessment bundle

Commands return non-zero on invalid input or an unsupported evidence claim. They are local development interfaces, not stable production APIs.

12. Proposed Milestone 5 data types

The next implementation should introduce separately versioned contracts:

Enterprise catalogue record

- `asset_id`, `service_id`, `owner_id`, `environment`
- business criticality, data classification, protection lifetime
- authoritative source, source version, effective time
- supplier/product/version and lifecycle dates
- dependency references and evidence status

Migration work item

- stable work ID and linked inventory/dependency IDs
- current and target cryptographic state by function/protocol
- wave, owner, approver, target date, readiness status
- blockers, prerequisites, rollback criteria, verification plan
- exception reference and decision history

Decision record

- decision ID, type, actor identity, role, time
- object and version decided upon
- rationale, evidence references, conditions, and expiry
- cryptographic audit receipt where required

Verification record

- approved target and change reference
- pre-change and post-change observations
- interoperability and negative-test results
- residual findings and regression checks
- verifier identity and independence level

13. Proposed migration state machine

```
discovered
  -> context-required
  -> assessed
  -> planned
  -> approval-required
  -> approved
  -> implementation-external
  -> verification-required
  -> verified

Any active state -> blocked
Any approval state -> excepted (time bounded)
Verified -> regressed (new contradictory evidence)
```

Transitions must be policy-checked, identity-bound, append-audited, and reject stale object versions.

14. Error-handling rules

- Contract errors: reject without partial downstream output.
- Replay conflicts: fail closed and preserve prior accepted evidence.
- Individual domain failures in a valid fixture page: store rejected outcome.
- Page envelope failure: no page state changes.
- Sink failure: report failure; production design must retry from an outbox.
- Unsupported evidence level: reject rather than downgrade silently.
- Missing business context: preserve exposure but withhold final priority.

15. Observability requirements for production

Metrics should include intake rate, schema failures, replay duplicates, conflicts, checkpoint lag, parser latency, sink backlog, assessment counts, unknown-field rates, evidence age, coverage gaps, workflow ageing, expired exceptions, and verification regressions. Logs must use correlation IDs and avoid leaking full certificate identities or infrastructure metadata by default.

Requirements Traceability Matrix

1. Purpose

This matrix links requirements to design, implementation, and executable verification. A row marked planned is a controlled gap, not evidence of completion.

2. Implemented and partial requirements

Requirement	Status	Design/code evidence	Test evidence
FR-001 Versioned validated observations	Implemented	`schemas/`, `SchemaRegistry`, `CertificateEvent`	`test_published_examples_conform_to_full_schemas`, `test_legacy_certificate_contract_requires_explicit_upgrade`
FR-002 Preserve provenance	Implemented for fixtures/lab	`CertificateEvent.provenance`, `CertificateTransparencyCollector`	`test_collects_pages_preserves_evidence_and_alerts`
FR-003 Reject changed-content replay	Implemented	`CollectorState.check_duplicate`	`test_replay_is_idempotent_and_emits_no_second_alert`, `test_source_identity_reuse_with_changed_content_fails_closed`
FR-004 Derive X.509 properties	Implemented in lab	`X509CertificateDiscovery`	`test_golden_generated_corpus`, `test_rsa_and_ecdsa_parameters_are_derived_from_certificate`
FR-005 Separate evidence domains	Partial	Optional event `transport`, inventory `transport_observation`	`test_loopback_probe_records_tls_without_inventing_key_establishment`
FR-006 Carry coverage limits	Implemented for certificate inventory	`PqcInventoryService._record`	`test_classical_certificate_is_quantum_vulnerable_and_high_priority`
FR-010 Stable linked inventory	Implemented	UUIDv5 record, observation ID	`test_assessment_is_deterministic`, CT collector tests
FR-011 Separate function and algorithm	Implemented for certificate uses	`cryptographic_use.function` and `.algorithm`	Inventory schema and inventory tests
FR-012 Identify classical exposure	Implemented	`QUANTUM_VULNERABLE`, assessment rules	`test_classical_certificate_is_quantum_vulnerable_and_high_priority`
FR-013 PQC/hybrid requires verification	Implemented	Assessment findings PQC-005/PQC-006	`test_pqc_algorithms_are_candidates_not_declared_safe`, `test_hybrid_label_requires_protocol_verification`
FR-014 Preserve unknown	Implemented	Assessment algorithm	`test_unknown_algorithm_never_becomes_quantum_safe`
FR-015 Block unsupported priority	Implemented	Missing-context and priority logic	`test_missing_business_context_blocks_prioritisation`
FR-030 Deterministic detection	Implemented	`CertificateMonitoringPipeline`	Pipeline test suite
FR-031 Evidence-linked explanation	Partial	Deterministic explanation and alert evidence	`test_unknown_domain_produces_high_severity_alert`
FR-032 AI cannot change decisions	Implemented structurally	No model call; explainer receives decided result	Determinism and alert schema tests
NFR-001 Fail closed	Implemented for local interfaces	Exact-key/domain/schema validation	Invalid fingerprint, timestamp, field, wildcard, file-size, and malformed-certificate tests
NFR-002 Auditability	Partial	Provenance, policy version, deterministic IDs, SQLite state	CT collection and deterministic-ID tests
NFR-006 Portable contracts	Implemented	JSON Schema Draft 2020-12	Full-schema tests
NFR-007 Reproducible assessment	Implemented	Deterministic rules	`test_assessment_is_deterministic`, `test_alert_identifier_is_deterministic`
NFR-008 Testability	Implemented for current scope	`tests/`	46 automated tests

3. Planned requirements and required proof

Requirement	Status	Required implementation evidence	Required acceptance evidence
FR-007 Multi-source collectors	Planned	Authenticated source adapters and conformance kit	Authorised integration tests; reconciliation and coverage report
FR-016 Enterprise correlation	Planned	Catalogue and dependency graph contracts	Known-estate fixture plus conflict/cycle tests
FR-020 Function-specific target state	Planned	Target-policy contract and compatibility rules	Approved scenario decisions and protocol tests
FR-021 Migration waves	Planned	Prioritisation and wave planner	Deterministic scenario outcomes and owner review
FR-022 Governed work items	Planned	Workflow state machine and RBAC	Positive/negative transition tests and audit export
FR-023 Exceptions	Planned	Expiry, approver, rationale, compensating controls	Expiry and unauthorised-approval tests
FR-024 Change approval/verification	Planned	Decision and verification records	Separation-of-duties and stale-version tests
FR-025 Regression detection	Planned	Post-change reconciliation	Controlled before/after scenario
FR-026 Coverage reporting	Planned	Authoritative scope denominator and status model	Reconciled pilot coverage sample
FR-033 Controlled delivery	Planned	Transactional outbox and authenticated adapter	Failure/retry/idempotency tests
NFR-003 Privacy	Planned beyond local use	Data classification, encryption, redaction, retention	Privacy review and access-control tests
NFR-004 Availability	Planned	Service SLOs, queueing, recovery design	Load, failover, restore, and game-day evidence
NFR-005 Isolation	Planned	Service identities and privilege boundaries	Architecture review and abuse-case tests
NFR-009 Scalability	Planned	Distributed deployment and partition model	Representative performance and capacity results
NFR-010 Accessibility	Planned	User research and accessible UI	Accessibility audit and task-completion study

4. Security invariant traceability

Invariant	Implementation	Evidence
Invalid input produces no alert	Validation precedes pipeline decision	Pipeline negative tests
Explanation cannot create/change alert	Decision calculated before explainer call	Pipeline structure and deterministic tests
DNS suffixes match label boundaries	`matches_policy`	Boundary and IDNA tests
Source identity cannot change bytes silently	Digest conflict check	CT provenance-conflict test
Unknown is never quantum-safe	Assessment decision order	Unknown and mixed-evidence tests
Observation/inventory/assessment commit together	SQLite transaction	CT persistence tests
Generated/fixture evidence cannot claim production	Evidence allowlist and reserved levels	Evidence escalation tests
TLS lab cannot reach external targets	Loopback IP validation	`test_loopback_probe_refuses_external_target`
Cipher/certificate does not invent KEM	Nullable transport field	Loopback TLS test

5. Traceability maintenance rule

A change is incomplete if it adds or materially changes a mandatory requirement without updating:

1. its identifier and acceptance criterion;
2. the HLD/LLD or an ADR;
3. its contract and compatibility statement;
4. automated or manual verification evidence;
5. the known limitation or risk entry where applicable.

Verification and Validation Plan

1. Objective

Verification asks whether the system was built according to its requirements and design. Validation asks whether it is suitable for the intended user and operational problem. The current repository provides substantial verification for a local reference slice and limited validation of the broader product need.

2. Test levels

Level	Purpose	Current evidence
Contract	Reject malformed or incompatible data	Full JSON Schema and domain tests
Unit	Verify normalisation, decisions, mappings, and failure rules	Python `unittest` suite
Component	Exercise collector, state, inventory, parser, and CLI boundaries	CT, inventory, discovery tests
Integration	Exercise multi-module local paths	Fixture-to-state/alert/inventory tests
Cryptographic artifact	Parse real generated PEM/DER certificates	Golden X.509 corpus
Protocol	Observe a real bounded TLS handshake	Loopback TLS test
Security negative	Demonstrate refusal and fail-closed properties	Oversize, malformed, replay conflict, evidence escalation, external target tests
System/operational	Validate deployed service behavior	Not yet available
User acceptance	Validate analyst and migration-owner workflows	Not yet available
Independent assurance	External review or penetration test	Not yet available

3. Current automated acceptance suite

The baseline has 46 tests covering:

- schema and format conformance;
- DNS policy normalisation and boundary behavior;
- deterministic alerts and assessments;
- invalid-input rejection;
- CT pagination, checkpoint restart, replay, provenance conflict, and size bounds;
- inventory classification, unknown handling, hybrid/PQC verification status,

missing business context, and function correctness;

- real generated RSA/ECDSA X.509 parsing;
- validity, CA, SAN/no-SAN, PEM/DER, malformed, and oversized cases;
- evidence-maturity enforcement;
- loopback TLS observation and non-loopback refusal;
- CLI output bundles.

Supported verification environments are Python 3.11-3.13 in project metadata and CI intent. The latest local evidence records complete passing runs on Python 3.12 and 3.13. A clean run should be attached to every release candidate.

4. Reproduction procedure

```
python3 -m venv .venv
.venv/bin/python -m pip install -e .
.venv/bin/python -m unittest discover -s tests -v
git diff --check
```

Record:

- commit SHA and branch;
- operating system and Python version;
- dependency resolution or lock evidence;
- start/end time;
- passed, failed, errored, and skipped counts;
- complete failure output;
- reviewer identity where evidence is used for a gate.

5. Golden-scenario validation

`examples/lab/scenarios.json` is the controlled expected-result manifest. Changes require review because a changed expectation can hide a regression. Golden scenarios must include at least:

- RSA 2048 valid end entity;
- ECDSA P-256 valid end entity;
- RSA 3072 expired certificate;
- ECDSA P-384 not-yet-valid certificate;
- RSA 4096 CA without SAN.

Future migration scenarios should add services, owners, data lifetimes, dependencies, target states, blockers, exceptions, and expected wave outcomes.

6. Production-readiness validation gates

Gate A: authenticated pilot intake

- Written authorisation and target scope.
- Collector identity bound to tenant and source.
- TLS and credential handling reviewed.
- Bounded collection, timeout, retry, and stop controls tested.
- Raw evidence protected and retention approved.
- Coverage denominator supplied by an authoritative source.

Gate B: inventory quality

- Sample records independently compared with source systems.
- False merge and missed merge rates measured.
- Unknown and stale evidence rates reported.
- Asset, owner, and dependency conflicts remain visible.
- No unsupported completeness claim.

Gate C: migration workflow

- Role and separation-of-duties matrix approved.
- State transitions and stale-write prevention tested.
- Exception expiry and escalation demonstrated.
- Before/after verification and regression paths exercised.
- Audit export reviewed by a non-developer stakeholder.

Gate D: production service

- Threat model and data-protection assessment updated.
- Dependency and container scanning completed.
- Penetration test findings resolved or accepted.
- Backup restore, failover, and incident runbooks exercised.
- Monitoring, on-call ownership, SLOs, and support model approved.

Gate E: optional execution

- Business need proves that integrated execution is preferable to an external

change system.

- Narrow adapter design and environment boundaries approved.
- Dual control, just-in-time access, dry run, rollback, and kill switch tested.
- Analysis and AI services have no access to execution credentials.
- Independent security approval obtained.

7. Test data strategy

Use data in increasing order of assurance:

1. Hand-authored simulated fixtures for rule edge cases.
2. Generated cryptographic artifacts for parser and protocol behavior.
3. Explicitly sanitised captures for realistic variation.
4. Authenticated production observations in an authorised pilot.
5. Independently verified samples for assurance measurement.

Never relabel a lower level. Sanitisation must preserve the property under test, remove secrets and unauthorised identifiers, document transformations, and retain source approval outside the repository.

8. Exit criteria for Milestone 5

1. Fictional enterprise catalogue and dependency graph contracts validate.
2. Migration work, decision, exception, and verification contracts validate.
3. Deterministic wave assignment handles required context and unknowns.
4. Cycles, conflicting ownership, unsupported targets, stale decisions, expired

exceptions, and missing verification fail safely.

1. Coverage and migration completion are reported separately.
2. Existing 46 tests remain passing and new requirements have tests.
3. ADR, threat model, PRD, LLD, and traceability matrix are updated.

Security, Risk and Assurance Plan

1. Assurance position

The current implementation demonstrates controlled engineering properties. It does not demonstrate production security, full discovery coverage, compliance, or successful migration of a real organisation.

Permitted current claims:

- versioned evidence and assessment contracts exist;
- deterministic reference rules are executable and tested;
- generated X.509 artifacts are parsed using a real cryptographic library;
- replay, conflict, evidence escalation, malformed input, and external target

refusal have automated tests;

- current limitations are documented.

Prohibited current claims:

- the product has discovered an enterprise estate;
- a customer or organisation is quantum-safe;
- the product validates complete TLS or PKI security;
- the product is certified, independently audited, or production hardened;
- its AI autonomously manages cryptography;
- its current evidence is authenticated production evidence.

2. Assets requiring protection

- Source observations and their provenance.
- Cryptographic inventory and dependency intelligence.
- Asset identities, owners, classifications, and data lifetimes.
- Policies, assessments, migration targets, and exception records.
- Approval identities and audit history.
- Future collector and execution credentials.
- Availability and integrity of migration-status reporting.

3. Primary threat scenarios

Scenario	Consequence	Current or required control
Forged or mislabelled evidence	Incorrect exposure or false completion	Strict contracts now; authenticated collector registry required
Missing source represented as complete coverage	Hidden vulnerable systems	Explicit coverage limits; authoritative denominator required
Correlation merges unrelated assets	Wrong owner or unsafe migration	Source lineage, conflict representation, review thresholds
Algorithm label represented as protocol proof	False PQC assurance	Function-specific evidence and verification-required states
Policy tampering	Suppressed work or manipulated priority	Version, approval, signature, effective date, rollback
AI prompt or evidence injection	Misleading advice or unsafe action	Treat output as untrusted; deterministic decisions; no credentials
Privilege convergence	One compromise can decide and execute	Separate collector, analysis, approval, and execution identities
Exception never expires	Permanent unmanaged exposure	Mandatory expiry, owner, approver, escalation, reporting
Sensitive inventory disclosure	High-value infrastructure intelligence leaks	Encryption, tenant isolation, field authorisation, redaction, audit
Dependency/supplier change	Verified item silently regresses	Continuous reconciliation and regression state
Parser or fixture resource abuse	Denial of service	Bounds now; sandbox, quota, and isolation required
Audit deletion or rewrite	Inability to prove decisions	Append-oriented records and tamper-evident external export

The detailed repository threat model remains the authoritative technical threat inventory for implemented milestones.

4. Control objectives

Evidence trust

- Authenticate collector workload and source account.
- Bind evidence to tenant, collector, scope, time, and software version.
- Use digest/signature semantics accurately; a hash is not authentication.
- Prevent an unknown collector from self-asserting a high assurance level.
- Retain raw evidence or a defensible reference and transformation history.

Decision integrity

- Version policy and record the exact version used.
- Make priority and transition gates deterministic.
- Require reasoned, named approval for exceptions and material changes.
- Prevent requesters from approving their own high-risk work.
- Reject decisions against stale object versions.

AI governance

- Limit AI to evidence correlation, summarisation, drafting, and options.
- Require structured outputs with source references and confidence.
- Evaluate hallucination, omission, injection, and inconsistent recommendation.
- Never give a model direct CA, HSM, KMS, trust-store, or deployment authority.
- Log accepted, rejected, and edited recommendations for evaluation.

Execution safety

- Prefer external change systems until integrated execution is justified.
- If added, isolate each adapter and environment.
- Use narrow, short-lived credentials and explicit approved change payloads.
- Provide dry-run, idempotency, rollback, rate limits, and a kill switch.
- Require independent post-change evidence before marking verified.

5. Risk register

ID	Risk	Likelihood	Impact	Current disposition
R-001	Lab success mistaken for production readiness	Medium	High	Mitigated by evidence levels and claim boundaries
R-002	Certificate discovery misses non-certificate cryptography	High	High	Accepted for current scope; multi-source roadmap required
R-003	Missing ownership blocks prioritisation	High	High	Explicitly blocks final priority; catalogue planned
R-004	Dependency graph is incomplete or stale	High	High	Planned reconciliation and confidence model
R-005	PQC support differs across protocol peers/products	High	High	Verification required; no algorithm-only completion
R-006	SQLite/JSONL used beyond reference scale	Medium	High	Architecture limitation; replace before production
R-007	Sensitive discovery data is exposed	Medium	High	Local-only now; privacy controls required before pilot
R-008	AI scope expands into ungoverned automation	Medium	Critical	Explicit authority boundary and future approval gate
R-009	Standards or supplier roadmaps change	High	Medium	Versioned policy, target state, and periodic review
R-010	Documentation drifts from code	Medium	Medium	Traceability and release evidence review

Risk owners, target dates, treatment cost, and residual ratings must be assigned before an authorised pilot.

6. Assurance artefacts by gate

Gate	Required evidence
Reference release	Source, schemas, tests, ADRs, threat model, traceability, reproducible run
Pilot approval	Scope/authorisation, data assessment, architecture review, operating model, rollback/stop procedure
Production approval	Secure build evidence, access model, pen test, recovery test, monitoring, incident plan, residual-risk acceptance
Migration completion	Approved target, change reference, interoperability tests, post-change discovery, independent verification where required

7. Compliance mapping approach

Do not claim compliance from this repository alone. When a target organisation and jurisdiction are known:

1. identify applicable obligations and internal control framework;
2. map control objectives to system capabilities and operating procedures;
3. record inherited, shared, and customer responsibilities;
4. attach objective evidence and test frequency;
5. identify gaps and approved remediation;
6. obtain qualified legal, compliance, and audit review where required.

8. Review cadence

- Every schema or security-boundary change: update LLD, traceability, and threat

model in the same change.

- Every release: reproduce tests and archive the evidence manifest.
- Quarterly during development: review risk register and standards assumptions.
- Before each pilot or new source: perform source-specific threat modelling.
- Before execution capability: conduct an independent architecture and security review.

Delivery Roadmap and Governance

1. Completed foundation

Milestone	Outcome	Evidence
1	Contract-first certificate monitoring vertical slice	Commit `bfdc4ff`, ADR-001, pipeline tests
2	Replay-safe bounded intake and durable evidence	Commit `20e5705`, ADR-002, CT tests
3	Cryptographic inventory and deterministic PQC exposure	Commit `887fdb3`, ADR-003, inventory tests
4	Controlled real X.509 and loopback TLS discovery lab	Commit `be71182`, ADR-004, discovery tests

These commits are local project evidence unless and until deliberately pushed and reviewed on GitHub.

2. Recommended roadmap

Milestone 5 - Migration management domain

Build a fictional enterprise dataset and the core governance contracts:

- catalogue, owner, service, supplier, and dependency records;
- current/target cryptographic state by function and protocol;
- migration work item, wave, decision, exception, and verification records;
- deterministic priority and wave prerequisites;
- coverage accounting and regression states;
- full traceability and tests.

Exit: a reviewer can move a fictional portfolio from discovery through approved plans and verified outcomes without production data or change authority.

Milestone 6 - Evidence reconciliation and offline bundle

- multi-observation correlation with conflict lineage;
- CBOM-style export and dependency graph views;
- sanitisation tool and customer-controlled offline evidence bundle;
- signed manifest, import validation, and data-quality report;
- report generation for technical and executive audiences.

Exit: realistic data can be exchanged and assessed without uncontrolled direct connectivity.

Milestone 7 - Authorised read-only pilot

- select one bounded source and obtain written authorisation;
- implement collector authentication and scope policy;
- add authoritative asset denominator and ownership enrichment;
- operate read-only, measure quality, and report unknowns;
- perform privacy and security review.

Exit: evidence-backed pilot report with measured, not assumed, coverage and error rates.

Milestone 8 - Production control-plane foundation

- tenant and identity architecture;
- durable stream, evidence storage, transactional workflow, and outbox;
- UI/API, RBAC, separation of duties, audit export;
- SLOs, recovery, observability, secure delivery pipeline;
- independent security testing.

Exit: approved production service for discovery and migration governance, still without autonomous cryptographic changes.

Milestone 9 - Optional controlled integrations

- integrate ticketing, CMDB, PKI, KMS/HSM, and deployment evidence;
- add narrowly scoped action proposals or external change triggers;
- consider isolated execution only after a separate risk decision.

Exit: verified end-to-end organisational workflow with explicit authority.

3. Governance roles

Role	Accountable decision
Product owner	Product outcomes, scope, and user acceptance
Security/cryptography authority	Algorithm policy and target-state approval
Enterprise architect	System and dependency architecture
Data owner	Collection purpose, classification, retention, and access
Service owner	Migration plan, implementation, and operational acceptance
Change authority	Production change approval and scheduling
Risk owner	Exception and residual-risk acceptance
Independent verifier	Evidence sufficiency for high-risk completion
Engineering team	Implementation, tests, operations, and defect resolution

One person may perform multiple roles in a prototype, but the production design must identify incompatible duties and compensating review.

4. Decision gates

Gate	Decision	Minimum evidence
G0 Concept	Continue problem discovery	Problem statement and stakeholder need
G1 Reference architecture	Accept design direction	PRD, HLD, threat model, ADRs
G2 Reference implementation	Accept technical feasibility	LLD, source, schemas, tests, evidence guide
G3 Pilot	Permit bounded real data	Authorisation, source scope, privacy/security review, stop plan
G4 Production	Operate as a service	SLOs, access model, recovery, monitoring, pen test, runbooks
G5 Migration execution	Permit change authority	Separation of duties, adapter review, dry run, rollback, independent approval

Passing one gate does not imply the next.

5. Delivery controls

- Use short-lived branches and reviewed pull requests.
- Require tests, schema compatibility review, threat-model delta, and traceability update for material changes.
- Pin external build actions and dependencies where practical.
- Produce a release evidence manifest with commit, environment, tests, schema versions, known issues, and artifact hashes.
- Sign releases and produce an SBOM before pilot distribution.
- Never mix secrets, customer evidence, or synthetic private keys into source control.

6. Backlog priority method

Prioritise features using:

1. assurance or safety risk reduced;
2. evidence needed for the next gate;
3. user value and decision enabled;
4. dependency order;
5. reversibility and learning value;
6. engineering and operating cost.

This places migration-domain modelling before a broad UI and authenticated read-only discovery before any execution authority.

7. Change control

Material changes require an ADR when they alter:

- trust boundaries or privilege;
- evidence or assurance semantics;
- canonical contract compatibility;
- data retention/classification;
- deterministic decision authority;
- workflow states and approval rules;
- production deployment or external integrations.

8. Immediate next backlog

1. Define Milestone 5 contracts and fictional enterprise scenarios.
2. Add a migration state-machine ADR.
3. Add catalogue, dependency, work-item, decision, exception, and verification schemas.

1. Implement deterministic prioritisation with explicit unknown/block states.
2. Add coverage and migration dashboards as generated local reports.
3. Extend the threat model and traceability matrix.
4. Run the full suite on Python 3.11, 3.12, and 3.13.

Project Evidence and Demonstration Guide

1. Claim being demonstrated

The repository is a working local reference implementation of an evidence-backed cryptographic discovery and PQC assessment foundation. It demonstrates deterministic certificate monitoring, replay-safe evidence intake, certificate-focused inventory and assessment, real generated X.509 parsing, and a bounded loopback TLS observation.

It does not demonstrate production estate coverage or completed PQC migration.

2. Evidence baseline

```
git log --oneline --decorate -5
git status --short --branch
git show --stat --oneline be71182
```

Expected milestone history:

```
be71182 Build controlled cryptographic discovery lab
887fdb3 Build PQC inventory and exposure foundation
20e5705 Build trustworthy CT intake milestone
bfdc4ff Build initial certificate monitoring control plane
```

If the current commit differs, record it and review the delta instead of assuming that this guide still describes the code exactly.

3. Environment setup

```
python3 --version
python3 -m venv .venv
.venv/bin/python -m pip install -e .
.venv/bin/python -m pip freeze
```

The repository currently pins `cryptography==50.0.2` and `jsonschema[format-nongpl]==4.26.0`.

4. Automated proof

```
.venv/bin/python -m unittest discover -s tests -v
git diff --check
```

Expected baseline: 46 passing tests. Do not hide failures or skips. A platform that prevents local socket creation may block the loopback test; run that test in an authorised local or CI environment and record the constraint separately.

5. Demonstration A - deterministic certificate alert

```
.venv/bin/python -m crypto_control_plane.cli \
  --event examples/events/unknown-domain-issued.json \
  --policy examples/policy.json \
  --output build/alerts.jsonl
```

Review:

- rule ID `CERT-001`;
- unmatched name evidence;
- policy ID/version;
- deterministic severity, score, and alert ID;
- advisory explanation authority;
- source evidence and provenance.

Then run the known-domain example and confirm it produces no alert.

6. Demonstration B - replay-safe evidence intake

```
.venv/bin/python -m crypto_control_plane.ct_cli \
  --fixture-dir examples/ct/pages \
  --policy examples/policy.json \
  --state build/ct-state.sqlite \
  --output build/ct-alerts.jsonl
```

Run it a second time. The same source records should be duplicates and should not produce duplicate alerts. The tests separately demonstrate failure when a source identity is reused with changed content.

7. Demonstration C - PQC inventory assessment

```
.venv/bin/python -m crypto_control_plane.inventory_cli \
  --event examples/events/known-domain-issued.json \
  --output build/pqc-inventory-assessment.json
```

Review:

- evidence and observation linkage;
- cryptographic function and algorithm;
- certificate signature as a separate algorithm field;
- `certificate-only` coverage limitation;
- quantum-vulnerable classification for classical public-key evidence;
- missing context and verification requirements.

8. Demonstration D - real X.509 parsing

Use the test suite's generated certificate corpus or generate a synthetic certificate through `SyntheticCertificateFactory`. Then run:

```
.venv/bin/python -m crypto_control_plane.discovery_lab_cli \
  --certificate /path/to/generated-certificate.pem \
  --evidence-level generated \
  --output build/x509-discovery-bundle.json
```

Review the derived fingerprint, public-key family/size or curve, signature algorithm/OID/hash, validity, SAN, key usage, EKU, inventory record, and PQC assessment. Confirm that an absolute input path is not leaked into evidence.

9. Demonstration E - security boundaries

The most efficient proof is to run the named tests:

```
.venv/bin/python -m unittest \
  tests.test_discovery_lab.DiscoveryLabTests.test_loopback_probe_refuses_external_target \
  tests.test_discovery_lab.DiscoveryLabTests.test_fixture_cannot_be_upgraded_to_production_evidenc
e \
  tests.test_discovery_lab.DiscoveryLabTests.test_unknown_collector_cannot_claim_production_eviden
ce \
  tests.test_ct_collector.CertificateTransparencyCollectorTests.test_source_identity_reuse_with_ch
anged_content_fails_closed
```

These show that the implementation refuses external TLS targets, unsupported evidence escalation, and provenance conflict.

10. Reviewer checklist

- [ ] Repository and commit identity recorded.
- [ ] Working tree state recorded.
- [ ] Dependency versions recorded.
- [ ] Complete tests executed and results retained.
- [ ] At least one positive and one no-alert scenario reproduced.
- [ ] Replay/idempotency behavior reproduced.
- [ ] Inventory and assessment output reviewed.
- [ ] Real X.509 parsing demonstrated.
- [ ] Security refusal cases demonstrated.
- [ ] PRD-to-test traceability sampled.
- [ ] Threat model and limitations reviewed.
- [ ] No production, certification, coverage, or migration-completion claim made

without additional evidence.

11. Evidence manifest template

```
Review date:
Reviewer:
Repository:
Branch:
Commit SHA:
Operating system:
Python version:
Dependency evidence:
Test result:
Skipped tests and reason:
Demonstrations completed:
Findings:
Known limitations accepted:
Decision: accept / accept with conditions / reject
Decision authority:
```

12. What proves the next stage

Milestone 5 should be considered proven only when fictional enterprise scenarios can demonstrate:

- discovery records linked to owners and services;
- dependency-aware planning;
- explicit current and target states by cryptographic function;
- deterministic wave and blocker decisions;
- time-bounded exceptions and approval history;
- before/after verification evidence;
- separate coverage and migration metrics;
- safe handling of unknowns, conflicts, cycles, expiry, and regression.

ADR-001: Contract-first deterministic certificate monitoring slice

- Status: accepted
- Date: 2026-10-11

Context

The target system may eventually ingest certificate events from ACME, cloud certificate managers, cert-manager, Certificate Transparency, TLS discovery, Vault, HSM-backed services, and AD CS. It may also provide AI-assisted analysis and governed lifecycle workflows. Building all integrations at once would hide trust assumptions and prevent early end-to-end verification.

Decision

Implement the first slice as a synchronous, dependency-light Python pipeline:

1. Accept a versioned canonical certificate event.
2. Perform strict domain validation and DNS-name normalization.
3. Enrich the event against an explicit organization DNS policy.
4. Evaluate deterministic rule `CERT-001` (unknown DNS namespace).
5. Create a deterministic alert ID and risk score.
6. Add a deterministic, advisory explanation from structured evidence.
7. Append the alert to a local JSONL development sink.
8. Provide a corresponding SOC playbook and automated tests.

The schemas are language-neutral JSON Schema Draft 2020-12 documents. Detection results do not depend on an LLM. No component can issue, renew, revoke, install, or delete certificates or keys.

Consequences

Positive:

- The first milestone is locally executable and auditable.
- Collector and SIEM adapters can evolve around stable contracts.
- Tests capture security-relevant normalization and policy behavior.
- A future model adapter can be evaluated without changing alert semantics.

Negative:

- The local validator implements the security-critical subset of the published

schemas; independent full JSON Schema conformance tooling is still required.

- JSONL lacks production delivery, integrity, retry, and access controls.
- DNS allowlists do not establish legal or operational domain ownership.
- The synchronous path is not a scale architecture.

Guardrails for later milestones

- Sign or otherwise authenticate collector-to-control-plane events.
- Define tenancy, replay, idempotency, retention, and redaction policies.
- Treat model output as untrusted advisory content; validate structured output.
- Require deterministic policy and appropriate human approval for high-impact cryptographic operations.
- Isolate any future executor from analysis services and use narrow, expiring, environment-scoped credentials.

ADR-002: Trustworthy intake contracts and replay-safe fixture collection

- Status: accepted
- Date: 2026-10-11

Context

Milestone 1 proved the canonical event-to-alert path with hand-authored events. It did not prove full schema conformance, evidence preservation, restart behavior, replay handling, or a collector boundary. Connecting directly to a live CT service at this point would combine unproven parsing, transport, pagination, trust, and operational behavior.

Decision

Implement a provider-neutral decoded CT fixture source behind a narrow source protocol. Validate every page, event, policy, and alert using JSON Schema Draft 2020-12 plus domain checks. Add a versioned provenance envelope containing the collector identity/version, source-record identity, raw-entry SHA-256, stated verification state, and source cursor.

Persist the following in SQLite:

- collector checkpoints;
- the canonical raw CT entry;
- payload digest;
- canonical event;
- accepted, alerted, or rejected outcome;
- generated alert or validation error; and
- processing time.

Treat a repeated source identity with the same digest as a duplicate. Treat the same source identity with a different digest as a provenance conflict and fail closed. Checkpoint only after a fully processed page. A restart resumes after the last completed page; a partial-page retry is safe because completed entries are idempotent.

Consequences

Positive:

- Tests exercise pagination, restart, replay, conflict, evidence, and alerting

without external network or credential dependencies.

- Raw evidence and policy versions are linked to detection outputs.
- A future live adapter has a small, testable contract.
- CI validates the supported Python versions with read-only permissions and

immutable action commit pins.

Negative and residual:

- A SHA-256 digest detects changed bytes but does not authenticate the source.
- The decoded fixture contract is not an implementation of CT wire protocols or

cryptographic proof verification.

- SQLite is a single-process reference store, not a distributed event system.
- Appending an alert to JSONL and recording SQLite state are not one atomic

transaction. A production integration needs a transactional outbox and explicit delivery semantics.

- Policy ownership remains illustrative rather than authoritative.

Rejected alternatives

- **Live CT first:** rejected because transport and parsing failures would mask

contract and replay defects.

- **In-memory deduplication:** rejected because it does not survive restart.
- **Digest described as a signature:** rejected because an unauthenticated hash

provides integrity comparison, not source authenticity.

- **LLM extraction of CT records:** rejected because parsing and detection must

remain deterministic and auditable.

ADR-003: Evidence-backed cryptographic inventory and PQC exposure

- Status: accepted
- Date: 2026-10-11

Context

Milestones 1 and 2 proved deterministic certificate alerting and replay-safe evidence intake. The product goal is broader: an evidence-backed cryptographic discovery and PQC migration control plane. Certificate observations are useful discovery evidence, but they do not establish negotiated key establishment, application implementations, business ownership, dependency paths, or complete cryptographic coverage.

Decision

Introduce two separately versioned and validated artifacts:

1 A cryptographic inventory record that preserves the source observation,

cryptographic function, algorithm details, asset context, evidence confidence, and explicit coverage limitations.

1 A deterministic PQC assessment that classifies known classical public-key

exposure, marks unknown evidence as unknown, treats PQC algorithms as candidates requiring deployment verification, and refuses to finalise migration priority without required business context.

Extend the canonical certificate event to version 1.2 with decoded public-key, certificate-signature, key-usage, extended-key-usage, and optional business context. Persist validated inventory records and assessments atomically with the source processing outcome.

Safety semantics

- RSA, elliptic-curve, and finite-field DH families are classified as

quantum-vulnerable evidence.

- ML-KEM, ML-DSA, and SLH-DSA evidence is `candidate-quantum-resistant`, not a

declaration of system safety or migration completion.

- A hybrid label is `hybrid-unverified` until the protocol, implementation,

peers, trust chain, and fallback behavior are tested.

- Known vulnerable evidence dominates unknown evidence; an unknown field cannot

hide a known classical exposure.

- Certificate key usage indicates permitted capability, not proof of runtime

protocol negotiation.

- Missing evidence stays explicit and blocks unsupported conclusions.

Consequences

Positive:

- The same intake foundation now produces auditable discovery and assessment artifacts for the future migration control plane.
- Business context and algorithm exposure are separated, allowing exposure to be identified without inventing migration priority.
- Coverage limitations travel with each record and reduce false confidence.
- Later TLS, code, cloud, KMS, HSM, private-PKI, and vendor collectors can target the same inventory boundary.

Negative and residual:

- Algorithm values are supplied by decoded fixtures; live X.509 parsing and independent verification are not implemented.
- The current inventory covers certificates only and contains no resolved dependency graph.
- CT observations generally lack authoritative owner, service, data-lifetime, implementation, and supplier context.
- SQLite remains a single-process reference store.
- The version 1.2 event contract intentionally requires producers to provide explicit cryptographic evidence or `unknown`; version 1.1 events fail closed.

Rejected alternatives

- **Treat missing fields as safe:** rejected because absence of evidence is not evidence of quantum resistance.
- **Declare ML-DSA/ML-KEM observations compliant:** rejected because algorithms alone do not establish protocol, chain, implementation, or relying-party readiness.
- **Infer TLS key establishment from the certificate:** rejected because certificate authentication/signatures and negotiated key establishment are distinct evidence domains.
- **Build migration orchestration now:** rejected until discovery coverage, authoritative ownership, policy, approvals, separation of duties, audit, and rollback controls exist.

ADR-004: Controlled discovery lab without customer data

- Status: accepted
- Date: 2026-10-11

Context

The project has no authorised customer or production cryptographic dataset. Milestone 3 established inventory and PQC assessment contracts, but decoded fixtures do not validate real certificate parsing or TLS observation. Pausing would delay reusable engineering; treating synthetic evidence as production would create false assurance.

Decision

Build a controlled discovery lab with three evidence boundaries:

1 Generate real RSA and ECDSA X.509 artifacts for valid, expired,

future-validity, CA, end-entity, SAN, and no-SAN scenarios.

1 Parse PEM and DER artifacts with a pinned PyCA `cryptography` dependency,

deriving rather than accepting certificate algorithm metadata.

1 Perform a real TLS handshake only against loopback IP literals. Record the

TLS version and cipher suite while leaving negotiated key establishment null unless a future protocol-specific collector can evidence it directly.

Add an evidence-maturity field to every event, inventory record, assessment, and alert. Known collectors have allowed levels. Fixtures are simulation-only; the file parser accepts generated or sanitised captures; the loopback probe is generated evidence only. Authenticated-production and independently-verified claims are represented in the portable schema but rejected at runtime until an authenticated collector trust registry exists.

Because these fields change required contracts, Milestone 4 explicitly advances the certificate event to v1.3, the inventory and assessment contracts to v1.1, and the detection alert to v1.2.

Consequences

Positive:

- Real cryptographic artifacts now exercise parsing and classification.
- The project can progress without confidential production data.
- Evidence quality and claim boundaries are machine-readable.
- A real local handshake tests certificate-versus-transport separation.
- The probe cannot be repurposed to scan external addresses.

Negative and residual:

- Generated certificates do not reproduce vendor, scale, topology, private-PKI,

HSM, KMS, application-library, or operational edge cases.

- Self-signed lab certificates do not establish trust-chain behavior.
- The standard Python TLS interface used here does not directly expose the

negotiated key-establishment group, so that field remains unknown.

- Private keys are unencrypted because the local TLS server needs them; they are synthetic test material and must never be reused or imported into a production key store.
- Production coverage and interoperability remain unverified.

Rejected alternatives

- **Wait for customer data:** rejected because parsers, evidence contracts, safety checks, and migration workflows are independently testable.
- **Scan public or enterprise targets:** rejected because the milestone has no target authorisation and must not add an unrestricted network scanner.
- **Infer KEM from TLS cipher or certificate:** rejected because that would conflate distinct protocol evidence.
- **Label generated artifacts as verified production:** rejected because it would invalidate the evidence-backed product claim.

Threat model: Milestones 1 to 4

Assets

- Integrity and provenance of certificate observations
- Organization DNS policy and ownership context
- Alert integrity, severity, evidence, and audit trail
- Collector checkpoints, idempotency records, and preserved raw observations
- Cryptographic inventory records, algorithm evidence, and PQC assessments
- Asset ownership, business criticality, classification, and protection lifetime
- Certificate metadata that may expose internal names or infrastructure
- Future credentials and cryptographic workflow authority (not present today)
- Evidence-maturity labels and the boundary between lab and production claims
- Local TLS target selection and parsed certificate artifacts

Trust boundaries



Event, CT fixture, policy, SQLite, and JSONL files can be modified by a local actor. The digest supports comparison but is not a signature or tamper-evident audit record.

Threats and current mitigations

Threat	Impact	Current mitigation	Residual / future control
Forged event	False alert or concealed issuance	Strict shape/type/time validation; evidence fields retained	Authenticated collectors and signed provenance
Source ID reused with changed content	Evidence substitution	Stored digest comparison fails closed	Source authentication and tamper-evident log
Duplicate/replayed observation	Alert flooding	Durable source identity/digest idempotency	Distributed idempotency key and retention policy
Checkpoint loss or cycle	Missed data or infinite collection	SQLite checkpoints, page-completion updates, cursor-cycle and page limits	Replicated state, monitoring, recovery procedure
Policy tampering	Suppressed or flooded alerts	Policy is explicit and fail-closed on invalid data	Signed/versioned policy, change approval, independent ownership source
DNS normalization bypass	Unknown name treated as trusted	Lowercase, trailing-dot removal, IDNA normalization, wildcard restrictions, label-boundary suffix matching	Differential tests with production parser and public-suffix awareness
Replay/duplicate event	Alert flooding	Deterministic alert identifier	Durable deduplication store and replay window
Prompt/model manipulation	Misleading response or unsafe action	No model call; explanation cannot change detection/severity; no executor	Structured output validation, evaluation, isolation, human review
Sensitive-name leakage	Infrastructure disclosure	Local-only sink and minimal fields	Classification, redaction, encryption, tenant isolation, retention policy
Malicious JSON volume	Resource exhaustion	Bounded local CLI use only	Payload/field limits, streaming parser, quotas, backpressure
JSONL injection	Corrupted downstream parsing	One encoded JSON object per line	Authenticated structured transport and schema enforcement
Privilege creep	Unauthorized issuance/revocation/key action	No credentials, network calls, or action interfaces	Separate executor, least privilege, approvals, audit, rollback
Algorithm-label spoofing	False PQC-readiness claim	Closed algorithm family enum, OID shape validation, provenance retained	Parse signed source artifacts; independently verify implementations and chains
Missing coverage presented as safety	Unmanaged vulnerable cryptography remains hidden	Explicit `certificate-only` coverage and limitations; unknown stays unknown	Coverage metrics across network, code, cloud, PKI, KMS, and HSM collectors
Business-context poisoning	Incorrect migration order	Evidence fields remain separate from deterministic classification	Authoritative CMDB/owner sources, change approval, conflict handling
PQC or hybrid label treated as deployment proof	Premature completion claim	Status is `candidate-quantum-resistant` or `hybrid-unverified`, never safe/compliant	Protocol interoperability, chain, implementation, and relying-party verification
Sensitive inventory disclosure	Attacker gains infrastructure and dependency intelligence	Local-only reference storage	Encryption, tenant isolation, field-level authorization, audit, retention controls
Fixture promoted to production evidence	Test result presented as customer assurance	Known collectors have allowed evidence levels; fixture sources are simulation-only	Authenticated collector identity and signed evidence envelopes
Local TLS probe used for network scanning	Unauthorised access or SSRF-style reachability	IP literals only; loopback addresses only; bounded timeout; no redirect or credential support	Separate reviewed production collector with explicit scope and egress policy
Certificate parser resource abuse	Memory/CPU exhaustion or parser exploitation	One-megabyte input bound, pinned parser dependency, malformed input fails closed	Sandboxed parser service, quotas, dependency monitoring
TLS cipher label used as KEM proof	False conclusion about key establishment	Negotiated key establishment is explicitly `null` unless directly evidenced	Protocol-specific telemetry and independent handshake verification
Synthetic private key leakage	Lab key reused or mistaken for production material	Factory returns ephemeral, unencrypted keys only for local tests; artifacts labelled generated	Ephemeral storage, restrictive file mode, automated cleanup, never import into production stores

Security invariants

1. Invalid events or policy fail closed and produce no alert output.
2. Explanations cannot create alerts or change rule results, severity, or score.
3. DNS suffix matching occurs only at label boundaries.
4. Alert identifiers are stable for the same event and rule.
5. The pipeline performs no network or cryptographic lifecycle side effects.
6. The same collector/source identity cannot be accepted with a different raw payload digest.

1. Checkpoints advance only after all entries on a validated page are recorded.
2. Missing algorithm evidence is `unknown`, never quantum-resistant by default.
3. PQC and hybrid observations require deployment verification and are not labelled compliant or safe solely from algorithm names.

1. Certificate evidence does not prove negotiated key establishment or complete enterprise cryptographic coverage.

1. Observation, inventory record, and PQC assessment are committed atomically.
2. Simulation and generated evidence cannot claim authenticated-production or independently-verified status; both high-assurance levels are runtime- disabled until an authenticated collector trust registry exists.

1. The lab TLS probe cannot connect to a non-loopback target.
2. Cipher-suite and certificate observations do not invent negotiated key-establishment evidence.

Out of scope

Production authentication/authorization, multi-tenancy, service availability, live X.509 parsing, CA validation, CT wire parsing/proof verification, source authentication, distributed event delivery, transactional SIEM outbox, production SIEM authentication, authoritative CMDB enrichment, dependency-graph resolution, Internet or enterprise-network scanning, authenticated production discovery, non-certificate discovery, and remediation/orchestration.

SOC playbook: CERT-001 - certificate for an unknown DNS namespace

Purpose

Investigate a certificate issuance event containing at least one DNS name that does not match the approved organization namespace policy. A detection is not proof of malicious issuance or domain ownership.

Initial triage

1 Confirm the alert evidence, event identifier, source, observation time,

certificate fingerprint, issuer, subject, and unmatched DNS names.

1 Validate the source independently. For public certificates, inspect the

relevant Certificate Transparency evidence and certificate chain; do not rely solely on the alert explanation.

1 Check authoritative domain/asset ownership records, recent acquisitions,

approved third-party services, and pending DNS or certificate requests.

1 Search for related certificates, DNS changes, TLS endpoints, account

activity, and issuance events using the fingerprint, serial number, issuer, subject, and DNS names.

1 Determine whether the policy is stale, the event is malformed, the name is a

legitimate external dependency, or the issuance is unauthorized.

Escalation

- Escalate immediately for confirmed organization names issued by an unexpected

CA, active impersonation, exposed services, compromised accounts, or evidence of unauthorized control.

- Engage PKI, DNS, cloud, identity, legal/brand, and incident-response owners as

applicable.

Containment and recovery

Do not revoke, replace, block, or modify DNS solely because this rule fired. Use established incident and change procedures. If unauthorized issuance is confirmed, coordinate CA/account containment, revocation where applicable, credential rotation, DNS/service remediation, monitoring, and evidence preservation with the responsible owners.

Closure evidence

Record the ownership decision, authoritative sources checked, policy correction if any, related indicators, actions/approvals, residual risk, and monitoring or follow-up requirements.